

Q1-01-B

Give an algorithm to compute x^{20} that only uses multiplications and uses as few multiplications as possible. How many multiplications does your algorithm use?

Q1-02-B

Convert from base 10 to base 2: (a) $(10.125)_{10}$, (b) $(5.15)_{10}$.

Q1-03-B

Convert from base 2 to base 10: (a) $(1010.11)_2$, (b) $(1.101)_2 \times 2^2$.